

Q.1:- What are the different protocols you observe at the following layers of the protocol stack?

Ans:- We have to capture total 19020 packets in 229 sec.

A. Application layer:-

In these packets, we get these application layer protocol i.e.,

- i) DNS :- Displayed 227, about 1.2%.
- ii) HTTP :- Displayed 9, about 0.04%.
- iii) DHCP :- Displayed 59, about 0.3%.
- iv) SSDP :- Displayed 286, about 1.5%.

B. Transport layer:-

In these packets, we get these transport layer protocol i.e;

- i) TCP :- Displayed 2941, about 15.5%.
- ii) UDP :- Displayed 8262, about 43.4%.

C. Network layer:-

In these packets, we get these network layer protocol i.e,

- i) ICMPv4 :- Displayed 31, about 0.2%.
- ii) ICMP :- Displayed 23, about 0.1%.
- iii) IPv6 :- Displayed 2008, about 10.6%.
- iv) ARP :- Displayed 7647, about 40.2%.

② What is the total amount of data being received for the following two cases?

A) When you access `http://neverssl.com/`
IP addr = [34.223.124.45].

Total amount of data being received be :- 36 packets
with 825 bytes of data.

B) When you access `https://www.cornell.edu`
IP addr = [128.253.173.24]

Total amount of data being received from 9680 packet
with 823 bytes of data.

③ Answer the following when you access the site `http://neverssl.com/`.

A) How many HTTP GET requests do you observe? List down the GET requests.

Ans:- There are 5 HTTP GET requests, I observe.
These are following :-

1) 208 GET / connecttest.txt HTTP/1.1

2) 292 GET / pkiops/crl/Microsoft_E1.20ID%20Verified%20Code%20Signing%20PCA%202021.crl HTTP/1.1

3) 598 GET / online HTTP/1.1

4) 599 GET / online / HTTP/1.1

5) 511 GET / favicon.ico HTTP/1.1.

(B) What version of HTTP the server is running?

Ans: HTTP/1.1.

(C) For each of the HTTP GET requests, find out

i) The total number of TCP segments being received.
Ans: - 3822 TCP segments being received.

ii) The total amount of data being received in the corresponding HTTP response message.

Ans: - Total amount of data being received in the corresponding HTTP response message is 7978.

(D) Identify TCP three way handshake used to establish the connection with <http://neverssl.com>. Specify SYN, SYN-ACK, ACK packets and write the sequence number for each of the three messages.

Ans: - Get request

No →

→ GET /online HTTP/1.1.

stream no → 81

No.	info	Sequence no.
7539	[SYN]	0
7894	[SYN; ACK]	0
7895	[ACK]	1